

Kasra Davoodi

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Electrical Engineering & Information Technology
Karlsruhe, Germany

Pages: [Webpage](#), [LinkedIn](#), [Google Scholar](#)

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RESEARCH INTERESTS

- Machine Learning
- Autonomous System
- Robotics
- Computer Vision
- Healthcare AI
- Signal Processing

EDUCATION

K. N. Toosi University of Technology

B. Sc. in Electrical Engineering

Tehran, Iran

2020 – 2025

GPA: 3.84/4 (local GPA: 18/20)

Bachelor Thesis: "Segmentation of ICH in Brain CT Scans using a two-step deep learning system"

Supervisor: Prof. [Amirhossein Nikoofard](#)

RESEARCH EXPERIENCE

K. N. Toosi University of Technology

Research Assistant (Prof. [Amirhossein Nikoofard](#), Prof. [Mahdi Aliyari](#) at APAC Research Group)

Tehran, Iran

Oct 2023 – Oct 2025

- **Project: Classification and Segmentation of ICH in Brain CT scans.**
 - Developed a two-step lesion segmentation system.
 - Developed advanced training pipelines using PyTorch.
 - Collected HEMORICA dataset through cooperation with clinics.

K. N. Toosi University of Technology

Research Assistant (Prof. [Hossein Hosseini-Nejad](#), at [Zistel company](#))

Tehran, Iran

Dec 2022 - Sep 2023

- **Project: Clinical Pulse Oximeter Performance Enhancement.**
 - Developed a four-step algorithm for SPO2 and Heart rate assessment in PPG Signal.
 - Implemented the algorithm on nRF and STM32 microcontrollers.
 - Optimized speed and accuracy of Pulse Oximeter device.

INDUSTRIAL EXPERIENCE

Parto Dadeh Company:

Intern (under the supervision of Mr. Arash Malek and Mr. Kaveh Manafi)

Project: Software Design for a Telecommunication Device

- Developed backend and frontend of an application using QT framework.
- Applied OOP principles and C++ libs for software development.

(NOTE: I encourage you to explore my [Webpage \(click\)](#) for detailed technical information of my Publications and projects.)

PUBLICATIONS

- **Conference Papers:**

1. Hoseyni, M., **Davoodi, K.**, Pakdaman, F., Aliyari Shoorehdeli, M., & Nikoofard, A. (2024). "Comprehensive Hyperparameter Tuning to Enhance Deep Learning Performance for Intracranial Hemorrhage Classification in Head CT Scans." In *Proceedings of the Iranian International Conference on Biomedical Engineering (ICBME)*, Tehran, Iran. [Published at IEEE \(click for credential\)](#)

- **Journal Papers:**

2. Ghafari, Z., **Davoodi, K.**, Katoozian, D., & Hosseini-Nejad, H. (2025). "An Innovative Approach for Beat Detection and Quality Assessment in Photoplethysmography (PPG) Signals" (**Revision – Nature Scientific Reports**).
3. Hasani, Z., Mahdavi Moghadam, M., Mohammadi, R., Shirmohammadi, Z., Nikoofard, A.*, Nikahd, E., & **Davoodi, K.** (2024). "A Deep Reinforcement Learning-Based Mechanism for Throughput Enhancement in Wireless Sensor Networks" [Published at Nature Scientific Reports \(click for credential\)](#)

- **In preparation / Archive:**

4. **Davoodi, K.**, Hoseyni, M., Khoramdel, J., Barati, R., Mortazavi, R., Nikoofard, A., Aliyari Shoorehdeli, M., & Hatam Parikhan, J. (2025). "Hemorica: A Comprehensive CT Scan Dataset for Automated Brain Hemorrhage Classification, Segmentation, and Detection" ([first version in arxiv](#)).

PROJECTS

Computer Vision:

- **ICH Brain CT Segmentation via Deep Learning**

Developed a two-step segmentation system using TinyViT and Unet as backbones. [APAC Research Group]

- **ICH Brain CT Classification via Deep Learning:** Tuned a generalizable set of hyper-parameters, enhancing ResNet, VGG16, DenseNet, and MobileNet, modeling the clinical scenarios. [APAC Research Group]

- **Hemorica Dataset.** Led preparation, cleaning, and evaluation of multi-label segmentation dataset of intracranial hemorrhage. [APAC Research Group]

- **Autonomous Driving – Vehicle & Human Detection:** Utilized YOLOv8s for live vehicle and human detection with preprocessing, augmentation, class balancing, weighted loss. [NeuroMatch Academy]

- **License Plate Recognition:** Designed a six-step educational project for a computer vision course as a TA, focusing on deep learning fundamentals for students, from labeling to CNN design [CV course]

- **Chessboard digitalization:** Developed a four-step method using CNN and YOLOv5 to map chessboard arrangements to digital images. [CV course]

Machine Learning:

- **Quality Assessment of PPG Signals:** Merging a XGBoost model with the peak/valley detection as a pre-process, eliminating low-quality segments from the signal. The whole package can be used as a preprocessing module for further applications. [Zistel Company]

- **Built ML models to predict bank customer behavior** (classification & regression). [Principles of Intelligent Systems course]

Signal Processing:

- **Heart Rate & SpO2 Detection:** Developed a lightweight, adaptive, 4-step peak/valley detection algorithm for medical grade calculation of heart rate & SpO2. Deployed the code on STM32 and nRF microcontrollers [Zistel Company]

Software:

- **Developed a telecommunication application** (backend + frontend) with C++ (QT framework), capable of multi-channel transmit and receive. [Parto Dadeh Company]

TECHNICAL & RESEARCH SKILLS

Programming skills	Computer Vision	Embedded systems	Research Skills
• Python, C, C++, Matlab • Pandas, Seaborn • QT, Cube MX • GIT, Jupyter notebook	• Image processing – OpenCV, PIL • Deep Learning – CNN, ViT, YOLO • advanced pipeline development • PyTorch, Scikit-learn, Hugging Face	• STM32, nRF, AVR • Filtering, Real Time Analysis • Algorithm Design • System Optimization	• Literature Review • Documentation • Task Management • Team Work

IELTS & COURSE CERTIFICATES

English: **IELTS Score = 8.0** (Listening = 9.0, speaking = 7.5, Reading = 8.5, writing = 7.0). [click for report form](#)
Deep Learning Course and Project (128 hours - Neuromatch Academy – July 2024). [View Credential](#)

TEACHING EXPERIENCE

- Head TA for “**Fundamentals of Computer Programming.**” Prof. [Behrooz Nasihatkon](#)
- TA for “**Fundamentals of Computer Vision.**” Prof. [Behrooz Nasihatkon](#)
- Head TA for “**Electronics 1.**” Prof. [Amir Masoud Sodagar](#)
- Head TA for “**Electronics 2.**” Prof. [Ebrahim Nadimi](#)
- TA for “**Electronics 1.**” Prof. [Amir Masoud Sodagar](#)
- TA for “**Fundamentals of Computer Programming.**” Prof. [Hamed Khanmirza](#)

VOLUNTARY WORK

Manager of Education Section in IEEE KNTU Student Branch

- Led the IEEE KNTU Education Section, organizing workshops, courses, and seminars.
- Received recognition by [Prof. Granpayeh](#) in recognition of exceptional performance and team leadership ([View certification of appreciation](#)).
- Created multiple part-time jobs for graduate instructors.
- Delivered both technical and soft skills training for undergraduates.